

Combining information from adaptive experiments

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Abstract

Combining evidence across experiments is a fundamental problem for both statistical theory and practice. Yet a complete framework of guidelines discussing how aggregation should be approached across differing design settings remains underdeveloped. The problem becomes especially complex under adaptive randomization, in particular when experiments interact through their designs. In such settings, the allocation mechanism could be ancillary or not for the likelihood and may have an impact on the appropriate inference to carry out. This paper addresses the problem of combining data from multiple (adaptive) experiments and discusses how to effectively aggregate information. We address allocation-adaptive and covariate-adjusted procedures, as well as response-adaptive and covariate-adjusted response-adaptive designs, showing how the nature of these different procedures impacts on the information sharing from an inferential perspective. We further examine independent and interacting experiments, including concurrent studies in stratified populations and characterize the asymptotic behaviour of estimators. The results provide a useful perspective on when information borrowing across adaptive experiments preserves statistical validity and when additional work is required.

Keywords: Asymptotic Inference, Borrowing, Comparative Experiments, Conditional and Unconditional Inference, Sequential Designs

1 Introduction

The problem of borrowing information across different experiments arises from the need to coherently combine statistical evidence from multiple studies conducted under similar, though not necessarily identical, conditions. Each experiment provides only partial information about the phenomenon of interest, which, when considered in isolation, may be insufficient for precise or stable inference. From a statistical perspective, borrowing information exploits the underlying similarity among experiments to improve inferential efficiency (Li et al. (2023); Gao et al. (2025)), for example by reducing the variability of estimators or increasing the power of statistical tests. While this process is potentially useful in applied settings such as clinical trials, where combining evidence can accelerate drug development and improve efficiency of inference (see, for example, Viele et al. (2014)), it involves a delicate trade-off between information gain and the risk of introducing bias when differences across experiments are not properly accounted for. In particular, heterogeneity among studies could arise from differences in experimental designs, target populations, or model assumptions that make the direct combination of information nontrivial.

Information borrowing can take different forms. Since the seminal paper of Pocock (1976), it has been understood as the incorporation of external data to augment the information available for the analysis of a current experiment. Classical approaches include complete or partial pooling, mixed-effects models, and meta-analytic techniques (for a review, see Kotecha et al. (2025)). Bayesian methods are particularly appealing in this context, as hierarchical models naturally enable information sharing across studies while accounting for variability at multiple levels. In this setting, dynamic approaches have been proposed to adaptively modulate the degree of information sharing according to the compatibility of data from different experiments, with the aim of preserving inferential validity even in the presence of small heterogeneities. Indeed, Bayesian methods for information borrowing have been widely studied in the literature (see, among others, Neuenschwander et al. (2010); Hobbs et al. (2012); Schmidli et al. (2014); Viele et al. (2014); Gravestock et al. (2017)). Information can also be borrowed within a single experiment, where distinct stages of the study serve as separate sources of evidence. This arises, for example, in adaptive clinical trial designs, where an initial stage with fixed randomization is followed by a response-adaptive phase, and final inference combines information accrued across stages (see, e.g., Lane and Flournoy (2012), Lane et al. (2014), Lane (2020)). Information may also be borrowed at the design stage of an experiment, such as in Aletti et al. (2024), where information is shared adaptively across strata during the trial to guide treatment allocation. In other settings, particularly in early-phase trials, borrowing is achieved by combining subgroup estimates with external control data to reduce uncertainty (see, e.g., Turner et al. (2022); Schwartz et al. (2025)).

This paper provides a general framework for sharing information across multiple experiments, highlighting the impact of potentially different natures of the designs generating the data, as well as possible dependencies/interactions among them. In particular, we (i) discuss when inference should be conditional on the realized allocation versus unconditional with respect to the design; (ii) analyze both independent

and interacting experiments - including concurrent and stratified settings - and characterize the asymptotic behavior of pooled estimators under these scenarios; (iii) study a selective borrowing scheme both for the design and the analysis of stratified experiments that aggregates information only among homogeneous sub-populations, describing its effect in terms of effective sample size and asymptotic variance. In this work, we focus on fixed sample size experiments; therefore, adaptation concerns only the allocation mechanism and does not involve stopping rules, interim analyses or sample size recalculation (these issues are briefly discussed in Section 6).

The structure of the article is as follows. Section 2 formalizes the model and Section 3 discusses the conditional versus unconditional inferential paradigm. Focusing on homogeneous populations, Section 4 deals with the combination of independent and dependent experiments, providing consistency results and central limit theorems for the pooled estimator. Section 5 extends the analysis to heterogeneous populations and develops an adaptive borrowing mechanism along with its asymptotic properties. Finally, Section 6 outlines some open problems.

2 Preliminaries

Consider a sequential adaptive experiment aimed at comparing the effects of $K \geq 2$ treatments, denoted by $T_1, \dots, T_K$, in a given population of interest $\mathcal{P}$. At each step a subject is randomized to one of the available treatments according to a given randomization rule and a scalar outcome is observed. Let $\boldsymbol{\delta}_i = (\delta_{i1}, \dots, \delta_{iK})^t$ be the treatment allocation of the i th statistical unit, with $\delta_{ik} = 1$ if this subject is assigned to treatment k (for $k = 1, \dots, K$) and 0 otherwise, where $\sum_{k=1}^K \delta_{ik} = 1$ for every i . Moreover, denoting by $Y_i(k)$ its potential outcome to treatment k , we assume $\{Y_i(k)\}_{i \in \mathbb{N}}$ iid random variables with common probability density/mass function $p_{\boldsymbol{\gamma}_k}(\cdot)$, where $\boldsymbol{\gamma}_k \in \Omega \subseteq \mathbb{R}^r$ is the unknown parameter vector. At each step only one of the potential outcomes will be observed, with $Y_i = \sum_{k=1}^K \delta_{ik} Y_i(k)$, and the goal is to make inference on $\boldsymbol{\gamma}^t = (\boldsymbol{\gamma}_1^t, \dots, \boldsymbol{\gamma}_K^t)$. From now on, let $\mathbf{1}_d$ and $\mathbf{0}_d$ be the d -dim vectors of ones and zeros, respectively, while $\{\mathbf{e}_1, \dots, \mathbf{e}_K\}$ denotes the canonical bases of $\mathbb{R}^K$. Let $\mathbf{N}(n) = (N_1(n), \dots, N_K(n))^t$ be the K -dim vector of treatment assignments after n steps, with $N_k(n) = \sum_{i=1}^n \delta_{ik}$, then $\boldsymbol{\pi}(n) = n^{-1} \mathbf{N}(n)$ is the vector of allocation proportions to the arms, where $\mathbf{1}_K^t \mathbf{N}(n) = n$ and $\mathbf{1}_K^t \boldsymbol{\pi}(n) = 1$ for every n . Moreover, we set $\mathbf{Y}_{(n)} = (Y_1, \dots, Y_n)^t$ and $\boldsymbol{\delta}_{(n)} = (\boldsymbol{\delta}_1, \dots, \boldsymbol{\delta}_n)^t$, so $\mathcal{F}_n = \sigma(\mathbf{Y}_{(n)}, \boldsymbol{\delta}_{(n)})$ denotes the filtration containing all the information collected up to the n th subject (with $\mathcal{F}_0$ the trivial σ -field).

Adaptive experiments belonging to this framework share some common properties (see, among others, Baldi Antognini and Giovagnoli (2015)); indeed, for every $i = 1, \dots, n$,

- P1: $\Pr(Y_i \mid \boldsymbol{\delta}_i, \mathcal{F}_{i-1}) = \Pr(Y_i \mid \boldsymbol{\delta}_i)$, the so-called conditional independence of the outcomes given the assignments;
- P2: $\Pr(Y_i \mid \boldsymbol{\delta}_i = \mathbf{e}_k) = p_{\boldsymbol{\gamma}_k}(Y_i)$, namely each observation contributes only in the estimation of the effect of the assigned treatment;
- P3: $\Pr(\boldsymbol{\delta}_i = \mathbf{e}_k \mid \mathcal{F}_{i-1}) = \phi_{ik}$ does not depend on $\boldsymbol{\gamma}$.

Here, ϕ_{ik} denotes the allocation probability of the i -th subject to the k -th treatment. For example, under complete randomization, $\phi_{ik} = K^{-1}$ for every $k = 1, \dots, K$ and $i \geq 1$. By representing the statistical models through $\mathbf{p}_\gamma = (p_{\gamma_1}, \dots, p_{\gamma_K})$ and the chosen design via $D_n = \{\Pr(\boldsymbol{\delta}_i = \mathbf{e}_k \mid \mathcal{F}_{i-1}), i = 1, \dots, n\}$, an adaptive experiment will be denoted throughout the paper by $E = \{\mathcal{P}; T_1, \dots, T_K; D_n; \mathbf{p}_\gamma; \mathcal{F}_n\}$.

Clearly, non-adaptive designs are a special case of the present framework with $\Pr(\boldsymbol{\delta}_i \mid \mathcal{F}_{i-1}) = \Pr(\boldsymbol{\delta}_i)$ a-priori determined (like, e.g., for complete randomization), while deterministic procedures can be encompassed by degenerating D_n into a sequence of Dirac measures in $\{\mathbf{e}_1, \dots, \mathbf{e}_K\}^n$. In general, in adaptive designs the sample size n could be prefixed or adapted to the accrued information $\mathcal{F}_n$, for instance for futility or efficacy in sequential monitoring. Throughout the paper, we only consider fixed sample sizes, so that the adaptation always refers to the allocation process.

3 Conditional vs unconditional inference

One important issue when drawing inference under adaptive designs is whether or not to condition on the observed allocations. Usually, if the adaptation process does not involve the outcomes, the design is treated as ancillary with respect to the parameter of interest, and inference is carried out conditionally on the design. Conversely, when the design is adapted to the outcomes, inference should be carried out unconditionally with respect to the observed allocations. In what follows, we discuss how the nature of the sequential allocation mechanism determines the appropriate inferential framework.

Under P1-P3 the likelihood of γ under the experiment E is (see, e.g., Baldi Antognini and Giovagnoli (2015) and Rosenberger and Lachin (2015))

$$L(\gamma; \mathbf{Y}_{(n)}, \boldsymbol{\delta}_{(n)}) = \prod_{i=1}^n \prod_{k=1}^K p_{\gamma_k}(Y_i)^{\delta_{ik}} \times \prod_{i=1}^n \prod_{k=1}^K \phi_{ik}^{\delta_{ik}} \propto \prod_{k=1}^K \prod_{i=1}^n p_{\gamma_k}(Y_i)^{\delta_{ik}}, \quad (1)$$

where proportionality holds since, from P3, each conditional distribution ϕ_{ik} does not depend on γ . Therefore, as is well known, the maximum likelihood estimator (MLE) $\hat{\gamma}_n$ of γ is the same as in the non-adaptive setting, namely when the design is not a random process depending at each step on previous observed outcomes and allocations. However, the correct inferential framework (and therefore the distribution of the MLE) depends on the nature of the adopted design.

If the chosen design is only adapted to previous allocations,

$$\Pr(\boldsymbol{\delta}_i \mid \mathcal{F}_{i-1}) = \Pr(\boldsymbol{\delta}_i \mid \boldsymbol{\delta}_{(i-1)}), \quad i = 1, \dots, n$$

then, from P3, $\boldsymbol{\delta}_{(n)}$ represents an *ancillary* statistic for the likelihood (namely it is non-informative with respect to the unknown model parameters) and we should argue conditionally on the observed allocations. For these designs, usually called Allocation-Adaptive (AA) procedures (Efron's biased coin design is a well-known example), inference should be performed by considering the allocations as prefixed and focusing on the distribution of the outcomes given the design, i.e. $\Pr(\mathbf{Y}_{(n)} \mid \boldsymbol{\delta}_{(n)})$. This

corresponds to assume, instead of (1),

$$L(\boldsymbol{\gamma}; \mathbf{Y}_{(n)} \mid \boldsymbol{\delta}_{(n)}) = \prod_{k=1}^K \prod_{i=1}^n p_{\gamma_k}(Y_i)^{\delta_{ik}}, \quad (2)$$

depending only on the sufficient statistics of the assumed model. For ancillary designs, (2) corresponds to the conditional likelihood, namely the likelihood associated with the conditional distribution of $\mathbf{Y}_{(n)}$ given $\boldsymbol{\delta}_{(n)}$, obtained by dividing the joint likelihood in (1) by the marginal distribution of $\boldsymbol{\delta}_{(n)}$. Whereas, when adaptation is informative (i.e., for non-ancillary designs), (2) is no longer proportional to the conditional likelihood.

For instance, in the case of binary outcomes $\gamma_k \in (0; 1)$ and $p_{\gamma_k}(Y_i) = \gamma_k^{Y_i}(1 - \gamma_k)^{1-Y_i}$ for $k = 1, \dots, K$, so that (2) becomes

$$L(\boldsymbol{\gamma}; \mathbf{Y}_{(n)} \mid \boldsymbol{\delta}_{(n)}) = \prod_{k=1}^K \prod_{i=1}^n \gamma_k^{\delta_{ik} Y_i} (1 - \gamma_k)^{\delta_{ik}(1-Y_i)} = \prod_{k=1}^K \gamma_k^{S_k(n)} (1 - \gamma_k)^{N_k(n) - S_k(n)},$$

where $S_k(n) = \sum_{i=1}^n \delta_{ik} Y_i$ denotes the number of successes in treatment k after n steps. Moreover, since $S_k(n)$ and $N_k(n)$ are permutation invariant, then every adaptive experiment leading to the same observed allocations and successes in the treatment groups induces the same (conditional) likelihood.

Under this setting, a crucial role is played by the conditional (average) Fisher Information Matrix (FIM) $\mathbf{M}(\boldsymbol{\gamma} \mid \boldsymbol{\delta}_{(n)})$, which represents (under the usual Cramer-Rao regularity conditions) the inverse of the asymptotic variance of the MLEs. From P2, the conditional FIM is block diagonal with $\mathbf{M}(\boldsymbol{\gamma} \mid \boldsymbol{\delta}_{(n)}) = \text{diag}(\mathbf{M}(\boldsymbol{\gamma}_k \mid \boldsymbol{\delta}_{(n)}))_{k=1, \dots, K}$, where the k th block (corresponding to treatment k) is the r -dim square matrix

$$\mathbf{M}(\boldsymbol{\gamma}_k \mid \boldsymbol{\delta}_{(n)}) = -\frac{1}{n} \sum_{i=1}^n \delta_{ik} \mathbb{E} \left(\frac{\partial^2 \ln p_{\gamma_k}(Y_i)}{\partial \gamma_{kl} \partial \gamma_{kh}} \right)_{l, h=1, \dots, r}$$

and the expectation is calculated with respect to the conditional distribution p_{γ_k} .

When the allocation process depends on previous responses

$$\Pr(\boldsymbol{\delta}_i \mid \mathcal{F}_{i-1}) = \Pr(\boldsymbol{\delta}_i \mid \mathbf{Y}_{(i-1)}, \boldsymbol{\delta}_{(i-1)}), \quad i = 1, \dots, n$$

then $\boldsymbol{\delta}_{(n)}$ is informative about the unknown parameters (i.e., it is non-ancillary) and we should argue unconditionally with respect to the observed design. Such a class of procedures is called Response-Adaptive (RA), where well-known examples in the literature are the sequential maximum likelihood design (Melfi and Page (2000)), the doubly adaptive biased coin (Eisele and Woodroffe (1995)) and the ERADE (Hu et al. (2009)). Thus, for RA designs the randomized nature of the design should be accounted for and inference should be performed by considering the joint distribution of the outcomes and the allocations $\Pr(\mathbf{Y}_{(n)}, \boldsymbol{\delta}_{(n)})$. However, from (1), the likelihood does not depend only on the sufficient statistics of the model and it is no longer

permutation invariant unless $\phi_{ik} = \phi_k$ for every i (i.e., if the allocation probabilities in each treatment groups are exactly the same at each step). Therefore, contrary to the case of conditional inference, if we argue unconditionally then two different adaptive experiments with same observed allocations and successes in the treatment groups do not induce the same likelihood, which generally depends on the entire allocation process. Due to this dependence structure that should be accounted for, unconditional inference is highly complex, particularly for finite samples, where exact results exist only for dichotomous responses (Yi 2013; Baas et al. 2025). Nevertheless, asymptotic inference is feasible provided that the design satisfies certain regularity conditions.

Assume now that, for each treatment $k = 1, \dots, K$, the distribution p_{γ_k} belongs to the regular exponential family, parameterized (without loss of generality) as $\gamma_k = (\theta_k; \zeta_k)$, where $\theta_k = \mathbb{E}[Y(k)]$ is the parameter of interest and ζ_k a nuisance parameter. We denote this family by $EF(\theta_k; \zeta_k)$. Then, the MLEs $\hat{\theta}_n = (\hat{\theta}_{1n}, \dots, \hat{\theta}_{Kn})^t$ of $\theta = (\theta_1, \dots, \theta_K)^t$ are the sample means, namely $\hat{\theta}_{kn} = N_k(n)^{-1} \sum_{i=1}^n \delta_{ik} Y_i$ for $k = 1, \dots, K$; moreover, let $\mathbf{V} = \text{diag}(\mathbb{V}(Y(k)))_{k=1, \dots, K}$ (where each variance of the potential outcomes lies in $\mathbb{R}^+$), the conditional FIM is $M(\theta \mid \delta_{(n)}) = \text{diag}(\pi(n)) \mathbf{V}^{-1}$, depending on the design only through $\pi(n)$; whereas, the unconditional FIM becomes

$$M(\theta) = \text{diag}(\mathbb{E}[\pi(n)]) \mathbf{V}^{-1}, \quad (3)$$

where

$$\mathbb{E}(\pi_k(n)) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}_i(\phi_{ik}) \quad (4)$$

and the expectation $\mathbb{E}_i$ is taken with respect to the joint distribution of $(\mathbf{Y}_{(i-1)}, \delta_{(i-1)})$, which clearly depends on the chosen design and is not permutation invariant. Therefore, while for conditional inference the design can be collapsed into the observed treatment sample sizes, for unconditional inference we should take into account the entire allocation process (not only the final percentage of assignments to the treatments).

Regardless of the nature of the chosen allocation procedure, if the chosen adaptive randomization guarantees that the allocation proportions converge to a deterministic target π^* with components in $(0, 1)$, then these approaches fall into a unified and simplified framework (Baldi Antognini and Giovagnoli 2015). Indeed, if

$$\lim_{n \rightarrow \infty} \pi(n) = \pi^* \in (0; 1)^K \quad a.s. \quad (5)$$

where π^* is a non-random quantity, then

$$\lim_{n \rightarrow \infty} \hat{\theta}_n = \theta \quad \text{and} \quad \lim_{n \rightarrow \infty} M(\theta \mid \delta_{(n)}) = \text{diag}(\pi^*) \mathbf{V}^{-1} \quad a.s.$$

and

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathbf{X} \sim N(\mathbf{0}_K; \text{diag}(\pi^*)^{-1} \mathbf{V}) \quad (6)$$

or, analogously, $\text{diag}(\mathbf{N}(n))^{1/2}(\hat{\theta}_n - \theta) \xrightarrow{d} \text{diag}(\pi^*)^{1/2} \mathbf{X} \sim N(\mathbf{0}_K; \mathbf{V})$.

Thus, under (5), the MLEs are still strongly consistent and asymptotically normally distributed. Moreover, by (3) the unconditional FIM is still block-diagonal and it converges to the same limiting matrix as that of the conditional FIM:

$$\lim_{n \rightarrow \infty} M(\boldsymbol{\theta}) = \text{diag}(\boldsymbol{\pi}^*) \mathbf{V}^{-1} \quad a.s.$$

which is the inverse of the asymptotic (unconditional) variance of the MLEs in (6). To provide further clarification of this result, notice that condition (5) implies that the percentage of allocations of each treatment degenerates asymptotically to a constant. In such a case, even if we argue unconditionally, the random vector $\mathbf{Z}(n) = \boldsymbol{\pi}(n) - \boldsymbol{\pi}^* \rightarrow \mathbf{0}_K$ a.s. as n tends to infinity; therefore, its asymptotic distribution does not depend on the parameter of interest, meaning that the statistic $\mathbf{Z}(n)$ is approximately ancillary (see Cox and Hinkley (1974), Lane (2020)) and inference on $\boldsymbol{\theta}$ should be made conditionally on $\mathbf{Z}(n)$ (i.e., $\boldsymbol{\pi}(n)$) in a large sample approach. Therefore, conditional and unconditional inference asymptotically coincide, provided that the chosen design satisfies condition (5) (which may fail for more complex adaptive mechanisms, e.g. with interim decisions characterized by informative stopping rules, as discussed in Section 6).

Although, from (5), the variability of the design vanishes asymptotically, this does not hold for small/moderate sample sizes, where adaptation can introduce additional bias to the MLE and lead to substantial losses in terms of power and control of type-I errors; this occurs particularly when the observed realization of the (random) design deviates substantially from its expected value in (4) (see Marschner (2021)): in such a case, as also pointed out in Lane (2020) and Lane (2022), conditional information tends to provide a more accurate approximation of the variance of the MLE. Moreover, a valid alternative to asymptotic likelihood-based inference relies on randomization tests, which have been proposed as a robust solution also suitable for small samples (for a review, see Proschan and Dodd (2019)). Different approaches to implementing randomization tests exist: exact tests guarantee type I error control without relying on asymptotic arguments, while Monte Carlo tests estimate the critical value via simulation at reduced computational cost, though at the risk of type I error rate inflation due to Monte Carlo error (Proschan and Dodd (2019), Wang et al. (2020), Baas et al. (2025)).

4 Combining adaptive experiments in a homogeneous population

In what follows we illustrate how to combine the information coming from J adaptive experiments sharing the same K treatments and a common statistical model for the outcomes with the same target population $\mathcal{P}$, namely

$$E^j = \left\{ \mathcal{P}; T_1, \dots, T_K; D_{n_j}^j; p_{\boldsymbol{\gamma}_k} \in EF(\boldsymbol{\theta}_k; \zeta_k) \text{ for } k = 1, \dots, K; \mathcal{F}_{n_j}^j \right\}, \quad j = 1, \dots, J.$$

The j -th experiment is based on the design $D_{n_j}^j$ with n_j observations and $\mathcal{F}_{n_j}^j = \sigma(\mathbf{Y}_{(n_j)}, \boldsymbol{\delta}_{(n_j)})$ denotes the corresponding filtration. Clearly the other experiments are performed using different designs, potentially distinct from $D_{n_j}^j$ and interacting with it. Now we will analyze the inferential properties of the overall experiment $\tilde{E}$ over $n = \sum_{j=1}^J n_j$ observations

$$\tilde{E} = \{\mathcal{P}; T_1, \dots, T_K; D_n; p_{\gamma_k} \in EF(\theta_k; \zeta_k), \quad k = 1, \dots, K; \mathcal{F}_n\},$$

where $D_n = (D_{n_1}^1, \dots, D_{n_J}^J)$ and

$$\mathcal{F}_n = \mathcal{F}_{n_1}^1 \vee \dots \vee \mathcal{F}_{n_J}^J = \sigma(\mathbf{Y}_{(n_1)}, \dots, \mathbf{Y}_{(n_J)}, \boldsymbol{\delta}_{(n_1)}, \dots, \boldsymbol{\delta}_{(n_J)}).$$

Obviously, if for every $j = 1, \dots, J$ the design of E^j is non-adaptive, then the information can be directly pooled, since the outcomes of all sub-experiments could be regarded as J independent sequences of iid observations. This is also true for AA rules or Covariate-adjusted procedures (where the allocation process depends on some pre-fixed covariates/prognostic factors that will be balanced during the trial), since from conditional inference the outcomes should be treated as J independent sequences of iid observations and the observed design as a pre-fixed realization. Moreover, in both of these scenarios it is often possible to derive inferential properties that are valid for any sample size. However, this is not generally true when at least one of the sub-experiments is RA.

To formalize how the sub-experiments may (or may not) share information in their evolution, we introduce the following definition of independence between the corresponding designs, that will be denoted by $\perp$ (i.e., $X \perp Y$ means that the random variables X and Y are independent; the extension to conditional independence given a filtration is straightforward).

Definition 1 Two designs $D_{n_j}^j$ and $D_{n_l}^l$ are independent if and only if, for every $i_j = 1, \dots, n_j$ and $i_l = 1, \dots, n_l$,

$$\boldsymbol{\delta}_{i_j} \perp \mathcal{F}_{i_l}^l \quad \text{and} \quad \boldsymbol{\delta}_{i_l} \perp \mathcal{F}_{i_j}^j. \quad (7)$$

Due to the conditional independence of the outcomes, two experiments E^j and E^l are independent if and only if their designs are mutually independent, which will be denoted in what follows by $D_{n_j}^j \perp D_{n_l}^l$.

Some examples of independent experiments are discussed below.

- Multi-stage experiments where the allocations of each stage are independent of the results of previous stages.
- An actual experiment matched with historical data: for example, $D_{n_1}^1$ is the design of a trial conducted in the past and $D_{n_2}^2$ is a new run of the same protocol carried out subsequently (e.g., due to a new recruitment period, a new set of sites, or a replication study). In this setting, $D_{n_2}^2$ is typically planned without using the

available data from $D_{n_1}^1$; nevertheless, one may still borrow information from both experiments at the analysis stage. This assumes that the reference population is the same across both experiments and that possible temporal trends in the outcome are negligible. Consequently, borrowing information is generally avoided when a substantial amount of time separates the studies.

- Stratified experiments with respect to some baseline covariates, where the evolution of each stratum is independent of those of the others; for instance, stratified clinical trials by medical centers. In this setting, one may combine information across strata to make inference for the target population, provided that no stratification effect could affect the outcome.

Conversely, examples of dependent designs include:

- multi-stage experiments where the future stages are planned on the basis of the information accrued in previous stages.
- Response-Adaptive procedures with an initial burn-in period: since RA designs often require initial estimates in order to be implemented, it is now common practice in the literature to start with an initial phase, usually performed via complete randomization or AA procedures like permuted blocks (see for a review [Rosenberger and Lachin \(2015\)](#); [Baldi Antognini and Giovagnoli \(2015\)](#)); here, $D_{n_1}^1$ corresponds to the burn-in phase, while $D_{n_2}^2$ is explicitly allowed to depend on $\mathcal{F}_{n_1}^1$ through the estimates of the unknown model parameters obtained at the end of the burn-in; recent contributions in the literature suggest increasing the size of the burn-in phase to avoid anomalous inferential behavior, such as type I error inflation (see, e.g., [Baldi Antognini et al. \(2018, 2022\)](#)).

4.1 Combining independent experiments in a homogeneous population

Consider now an experiment $\tilde{E}$ consisting of $J \geq 2$ independent sub-experiments $E^1, \dots, E^J$ on the same target population $\mathcal{P}$. Since $D_{n_j}^j \perp D_{n_l}^l$ for every pair $(j, l) \in \{1, \dots, J\}$, the overall likelihood could be naturally factorized into the J sources of information. Indeed, from (7), for every $i_1, \dots, i_J$

$$\Pr(\delta_{i_j} \mid \mathcal{F}_{i_1}^1, \dots, \mathcal{F}_{i_J}^J) = \Pr(\delta_{i_j} \mid \mathcal{F}_{i_j-1}^j),$$

so that

$$\begin{aligned}
L(\gamma; \mathbf{Y}_{(n_1)}, \boldsymbol{\delta}_{(n_1)}, \dots, \mathbf{Y}_{(n_J)}, \boldsymbol{\delta}_{(n_J)}) &= \\
&= \Pr(\mathbf{Y}_{(n_J)}, \boldsymbol{\delta}_{(n_J)}; \gamma \mid \mathbf{Y}_{(n_{J-1})}, \boldsymbol{\delta}_{(n_{J-1})}, \dots, \mathbf{Y}_{(n_1)}, \boldsymbol{\delta}_{(n_1)}) \times \dots \\
&\quad \times \Pr(\mathbf{Y}_{(n_2)}, \boldsymbol{\delta}_{(n_2)}; \gamma \mid \mathbf{Y}_{(n_1)}, \boldsymbol{\delta}_{(n_1)}) \times \Pr(\mathbf{Y}_{(n_1)}, \boldsymbol{\delta}_{(n_1)}; \gamma) \\
&= \prod_{i_J=1}^{n_J} p_\gamma(Y_{i_J} \mid \boldsymbol{\delta}_{i_J}, \mathcal{F}_{i_J-1}^J, \mathcal{F}_{i_1}^1, \dots, \mathcal{F}_{i_{J-1}}^{J-1}) \Pr(\boldsymbol{\delta}_{i_J} \mid \mathcal{F}_{i_J-1}^J, \mathcal{F}_{i_1}^1, \dots, \mathcal{F}_{i_{J-1}}^{J-1}) \times \dots \\
&\quad \times \prod_{i_1=1}^{n_1} p_\gamma(Y_{i_1} \mid \boldsymbol{\delta}_{i_1}) \Pr(\boldsymbol{\delta}_{i_1} \mid \mathcal{F}_{i_1-1}^1) = \prod_{j=1}^J \prod_{i_j=1}^{n_j} p_\gamma(Y_{i_j} \mid \boldsymbol{\delta}_{i_j}) \Pr(\boldsymbol{\delta}_{i_j} \mid \mathcal{F}_{i_j-1}^j) \\
&= \prod_{j=1}^J L(\gamma; \mathbf{Y}_{(n_j)}, \boldsymbol{\delta}_{(n_j)}) \propto \prod_{j=1}^J \prod_{k=1}^K \prod_{i_j=1}^{n_j} p_{\gamma_k}(Y_{i_j})^{\delta_{i_j k}}.
\end{aligned}$$

Clearly, the assignments of the statistical units to the J sub-experiments could be deterministic or random. Therefore, for multi-stage experiments we could argue as $i = 1, \dots, n_1, n_1 + 1, \dots, n_1 + n_2, n_1 + n_2 + 1, \dots, n$ or, more in general, by assigning an experimental indicator $\mathcal{S}$ to the n statistical units, where $\mathcal{S}_i = j$ if the i -th subject is assigned to E^j (for $i = 1, \dots, n$). For instance, $\mathcal{S}$ could denote the medical center in a coordinated multi-center clinical trial, where each sub-experiment shares a common population of interest (see Section 4.2); otherwise, $\mathcal{S}$ could be a function of one or more covariates, leading to a stratified experiment with J strata (see Section 5). In this latter case, the reference populations of the sub-experiments are not homogeneous; however, since i) the covariates are usually assumed to be an exogenous process of iid nature and ii) the covariate distribution is ancillary to the likelihood, all the present results could also be applied to Covariate-Adjusted Response-Adaptive (CARA) designs as well, provided that i) the allocation process of each stratum does not depend on the information accrued in the other strata and ii) no stratification effects affect the responses. Some suitable extensions for dependent experiments will be discussed in Section 5.2. Clearly, if n_j are random quantities, these derive from a non-informative adaptation process, so we should argue conditionally on their observed value.

Let now $\mathbf{N}(n_j)$ and $\boldsymbol{\pi}(n_j) = n_j^{-1} \mathbf{N}(n_j)$ denote the total amount and the proportion of assignments to the K treatments under the experiment $j = 1, \dots, J$, then according to the overall experiment $\mathbf{N}(n) = \sum_{j=1}^J \mathbf{N}(n_j)$ and $\boldsymbol{\pi}(n) = \sum_{j=1}^J \omega(n_j) \boldsymbol{\pi}(n_j)$, where $\omega(n_j) = n_j/n$ and $\sum_{j=1}^J \omega(n_j) = 1$. Thus, the MLE $\hat{\boldsymbol{\theta}}_n$ of the parameters of interest $\boldsymbol{\theta}$ is a linear combination of the MLEs $\hat{\boldsymbol{\theta}}_{n_j}$ s of the sub-experiments:

$$\hat{\boldsymbol{\theta}}_n = \sum_{j=1}^J \text{diag}(\boldsymbol{\rho}(n_j)) \hat{\boldsymbol{\theta}}_{n_j}, \quad (8)$$

where $\boldsymbol{\rho}(n_j) = (\rho_1(n_j), \dots, \rho_K(n_j))^t$, with $\rho_k(n_j) = N_k(n_j)/N_k(n)$ for every $k = 1, \dots, K$ (with $\sum_{j=1}^J \boldsymbol{\rho}(n_j) = \mathbf{1}_K$). Clearly, naive information borrowing through pooled estimators may perform poorly in terms of mean squared error in the presence

of heterogeneity among sub-populations (i.e., in the case of treatment-effect heterogeneity); such a setting is treated separately in Section 5, where a simple pooling is replaced by a Response-Adaptive borrowing scheme.

The following Corollary provides the asymptotic inference for the MLE of the combined experiment regardless of the nature (adaptive or non-adaptive) of the designs of the J independent sub-experiments. In order to describe the asymptotic distribution of the MLEs notice that, as n diverges, at least one of $n_1, \dots, n_J$ must diverge and from now on we assume that

$$\lim_{n \rightarrow \infty} \omega(n_j) = \omega_j^* \in [0, 1] \quad \text{for } j = 1, \dots, J \quad (9)$$

to account for a possible finite sample size in some sub-experiments.

Corollary 1 *If for every $j = 1, \dots, J$, $\pi(n_j) \rightarrow \pi_j^* \in (0, 1)^K$ a.s., where π_j^* s are non-random quantities, then as n tends to infinity*

$$\pi(n) \rightarrow \pi^* = \sum_{j=1}^J \omega_j^* \pi_j^* \quad \text{and} \quad \hat{\theta}_n \rightarrow \theta \quad \text{a.s.}$$

Moreover, for every $j = 1, \dots, J$, $\rho(n_j) \rightarrow \rho_j^* = \text{diag}(\pi^*)^{-1} \omega_j^* \pi_j^*$ a.s. and, from the independence of the sub-experiments, the asymptotic normality in (6) still holds since $\mathbf{X} = \sum_{j=1}^J \text{diag}(\rho_j^*) \mathbf{X}_j$, where $\mathbf{X}_1, \dots, \mathbf{X}_J$ are independent random vectors jointly normally distributed with

$$\mathbf{X}_j \sim N(\mathbf{0}_K, \text{diag}(\omega_j^* \pi_j^*)^{-1} \mathbf{V}), \quad j = 1, \dots, J.$$

Proof Suppose now that $\omega_j^* \neq \{0, 1\}$ for every $j = 1, \dots, J$, then from (9), $\pi(n) \rightarrow \pi^* \in (0, 1)^K$ a.s. and $\rho(n_j) \rightarrow \rho_j^* \in (0, 1)^K$ a.s. Therefore, from (8), $\hat{\theta}_n \rightarrow \sum_{j=1}^J \text{diag}(\rho_j^*) \theta = \theta$ a.s., since $\sum_{j=1}^J \text{diag}(\rho_j^*)$ is the K -dim identity matrix. Moreover,

$$\sqrt{n}(\hat{\theta}_n - \theta) = \sum_{j=1}^J \text{diag}(\rho(n_j)) \sqrt{n}(\hat{\theta}_{n_j} - \theta),$$

which converges in distribution to $\mathbf{X} = \sum_{j=1}^J \text{diag}(\rho_j^*) \mathbf{X}_j$, since $\sqrt{n}(\hat{\theta}_{n_j} - \theta) \xrightarrow{d} \mathbf{X}_j$ for every j and the independence of the sub-experiments, where the random vector $(\mathbf{X}_1, \dots, \mathbf{X}_J)^t$ is jointly normal with $\mathbf{X}_j \sim N(\mathbf{0}_K; \text{diag}(\omega_j^* \pi_j^*)^{-1} \mathbf{V})$ since

$$\mathbb{V}(\sqrt{n}(\hat{\theta}_{n_j} - \theta)) = n \text{diag}(\mathbf{N}(n_j))^{-1} \mathbf{V} = \omega(n_j)^{-1} \text{diag}(\pi(n_j))^{-1} \mathbf{V} \rightarrow \text{diag}(\omega_j^* \pi_j^*)^{-1} \mathbf{V} \text{ a.s.}$$

If one or more sub-experiments have a bounded sample size, say if $\omega_j^* = 0$, as n tends to infinity $\mathbf{N}(n_j)$ also admits a bounded limit; in such a case, asymptotically, all non-degenerate contributions stem from the other sub-experiments. $\square$

4.2 Combining dependent experiments in a homogeneous population

We now deal with the case of multiple dependent experiments that are conducted concurrently, i.e., simultaneously on the same reference population, comparing the

same set of treatments and sharing the same outcome model. As discussed in Section 4.1, this setting corresponds to the case of multi-center coordinated clinical trials. Suppose that $\tilde{E}$ consists of J dependent and concurrent sub-experiments where, for every $i_j = 1, \dots, n_j$, the treatment assignment probability depends both on the history of its own sub-experiment and on that of the other:

$$\Pr(\delta_{i_j} \mid \mathcal{F}_{i_j-1}^j, \mathcal{F}_{i_l}^l : l \neq j), \quad j = 1, \dots, J. \quad (10)$$

As previously mentioned, the combined experiment can be globally indexed by $\mathcal{S}$. For each statistical unit $i = 1, \dots, n$, we assume that this experimental indicator is chosen/observed prior to the treatment assignment, where $\mathcal{S}_1, \dots, \mathcal{S}_n$ is an iid sequence of random variables independent of both treatment assignments and responses, where $\Pr(\mathcal{S}_i)$ does not depend on γ . At each step i , the filtration of $\tilde{E}$ can be viewed as the standard filtration enriched by the sequence $\mathcal{S}_{(i)} = (\mathcal{S}_1, \dots, \mathcal{S}_i)^t$, namely $\mathcal{F}_i = \sigma(\mathbf{Y}_{(i)}, \delta_{(i)}, \mathcal{S}_{(i)})$, so that the treatment assignment mechanism in (10) becomes

$$\Pr(\delta_i \mid \mathcal{F}_{i-1}, \mathcal{S}_i), \quad \text{for any } i = 1, \dots, n. \quad (11)$$

In practice, this framework describes an (overall) experiment in which each sub-experiment transfers its information to a central coordinating unit, where at each step all the available information is accumulated and used to allocate the next patient (the more general setting of heterogeneous sub-populations is addressed in Section 5). Similarly to the independent case, the overall likelihood $L(\gamma; \mathbf{Y}_{(n)}, \delta_{(n)}, \mathcal{S}_{(n)})$ factorizes into the J sources of information, namely

$$\begin{aligned} & \prod_{i=1}^n p_\gamma(Y_i \mid \delta_i, \mathcal{S}_i, \mathcal{F}_{i-1}) \Pr(\delta_i \mid \mathcal{S}_i, \mathcal{F}_{i-1}) \Pr(\mathcal{S}_i \mid \mathcal{F}_{i-1}) \\ &= \prod_{i=1}^n p_\gamma(Y_i \mid \delta_i) \Pr(\delta_i \mid \mathcal{S}_i, \mathcal{F}_{i-1}) \Pr(\mathcal{S}_i) \\ &\propto \prod_{i=1}^n p_\gamma(Y_i \mid \delta_i) = \prod_{j=1}^J \prod_{k=1}^K \prod_{i_j=1}^{n_j} p_{\gamma_k}(Y_{i_j})^{\delta_{i_j k}}, \end{aligned}$$

since the process $\{\mathcal{S}_i\}_{i \geq 1}$ is ancillary, and therefore the MLE coincides with the one in (8). Assuming (9), the following result extends Corollary 1 to the case of dependent and concurrent sub-experiments.

Corollary 2 *If for any $j = 1, \dots, J$, $\pi(n_j) \rightarrow \pi_j^* \in (0, 1)^K$ a.s., where π_j^* s are non-random quantities, then as n tends to infinity,*

$$\pi(n) \rightarrow \pi^*, \quad \rho(n_j) \rightarrow \rho_j^* \quad \text{and} \quad \hat{\theta}_n \rightarrow \theta \quad \text{a.s.}$$

Moreover, $\mathbf{X} = \sum_{j=1}^J \text{diag}(\rho_j^*) \mathbf{X}_j$, but now $\mathbf{X}_1, \dots, \mathbf{X}_J$ are dependent random variables normally distributed only marginally with $\mathbf{X}_j \sim N(\mathbf{0}_K; \text{diag}(\omega_j^* \pi_j^*)^{-1} \mathbf{V})$ for $j =$

$1, \dots, J$. Hence, if there exists an index $j \in \{1, \dots, J\}$ such that $\omega_j^* = 1$, then $\mathbf{X} \sim N(\mathbf{0}_K; \text{diag}(\omega_j^* \pi_j^*)^{-1} \mathbf{V})$.

Proof The proof follows easily from Corollary 1. $\square$

In the presence of dependence across experiments, no general statement can be made about the asymptotic distribution of $\hat{\theta}_n$ (i.e., $\mathbf{X}$) unless just one of the sub-experiments has an unbounded sample size. The same situation occurs when the dependence structure between the designs tends to vanish asymptotically and in such a case α -mixing conditions could guarantee the joint normality of $(\mathbf{X}_1, \dots, \mathbf{X}_J)^t$ and so those of $\mathbf{X}$.

Remark 1 (Consecutive and dependent experiments) As a special case of Corollary 2, suppose $\tilde{E}$ consists of $J = 2$ dependent sub-experiments, such that E^1 precedes E^2 . This situation arises, for example, when a RA design is implemented after a burn-in phase. In such a case, n_1 is a pre-fixed and finite integer and the allocation probabilities under E^2 depend on the overall filtration generated by E^1 . The expression in (11) can be simplified as

$$\Pr(\delta_i | \mathcal{F}_{n_1}^1, \mathcal{F}_{i-1}^2), \quad \text{for any } i = n_1 + 1, \dots, n = n_1 + n_2,$$

since all the information accrued during E^1 is available when E^2 starts. Under this scenario, $\omega_2^* = 1 = 1 - \omega_1^*$ and, as n increases, $N(n_1)$ remains a.s. bounded and also $\pi(n_1)$ stabilizes to a finite constant (vector). Thus, by Corollary 2, if $\pi(n_2) \rightarrow \pi_2^* \in (0; 1)^K$ a.s., where π_2^* is a non-random quantity, then as n tends to infinity:

$$\pi(n) \rightarrow \pi_2^*, \quad \rho(n_2) \rightarrow \rho_2^* = \mathbf{1}_K \quad \text{and} \quad \hat{\theta}_n \rightarrow \theta \quad \text{a.s.}$$

and

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathbf{X} = \mathbf{X}_2 \sim N(\mathbf{0}_K; \text{diag}(\pi_2^*)^{-1} \mathbf{V}).$$

Thus, asymptotically, the MLE depends only on the contribution of E^2 since n_1 should be assumed finite. Note that in this case, unlike in Lane and Flournoy (2012) and Lin et al. (2020), stable convergence does not hold, since the influence of E^1 vanishes as the size of E^2 increases. Stable convergence would instead occur under designs in which the randomization mechanism of E^2 depends persistently on the information generated during E^1 , so that the contribution of the first experiment persists asymptotically.

5 Combining adaptive experiments in a heterogeneous population

Throughout this section we deal with concurrent experiments, each carried out on a different patient sub-population, naturally leading to a stratified experiment. The overall population $\mathcal{P}$ is partitioned into J non-overlapping sub-populations $\mathcal{P}_j, j = 1, \dots, J$, leading to a set of $J \geq 2$ experiments. Suppose that all the experiments involve the same set of K treatments, whose effects are allowed to vary across sub-populations. Let $Y(k, j) \sim p_{\gamma_{k,j}}$ denote the potential outcome that would be observed if treatment k was assigned to a patient in the j th sub-population (i.e., stratum), where

$p_{\gamma_{k,j}}$ follows a regular exponential family with $\gamma_{k,j} = (\theta_{k,j}; \zeta_k)$, where $\theta_{k,j} = \mathbb{E}[Y(k, j)]$ is the parameter of interest (potentially varying across sub-populations) and ζ_k a treatment-specific nuisance parameter.

The j -th stratum-specific experiment is

$$E^j = \left\{ \mathcal{P}_j; T_1, \dots, T_K; D_{n_j}^j; p_{\gamma_{k,j}} \in EF(\theta_{k,j}; \zeta_k) \text{ for } k = 1, \dots, K; \mathcal{F}_{n_j}^j \right\},$$

where $D_{n_j}^j$ denotes the design implemented within stratum j over n_j observations and $\mathcal{F}_{n_j}^j = \sigma(\mathbf{Y}_{(n_j)}, \boldsymbol{\delta}_{(n_j)})$ is the corresponding filtration. Since each sub-experiment E^j is implemented for comparing the same K treatments (although within stratum-specific effects potentially different), the overall experiment could be naturally defined by

$$\tilde{E} = \left\{ \mathcal{P} = \bigcup_{j=1}^J \mathcal{P}_j; T_1, \dots, T_K; D_n; p_{\gamma_{k,j}} \in EF(\theta_{k,j}; \zeta_k) \text{ for } k = 1, \dots, K; j = 1, \dots, J; \mathcal{F}_n \right\},$$

where $n = \sum_{j=1}^J n_j$, $D_n = (D_{n_1}^1, \dots, D_{n_J}^J)$ is the global design (with experiment-specific designs independent or interacting) and $\mathcal{F}_n = \mathcal{F}_{n_1}^1 \vee \dots \vee \mathcal{F}_{n_J}^J$ is the global filtration.

We assume an ANOVA-type set-up,

$$\theta_{k,j} = \theta_k + \alpha_{k,j}, \quad k = 1, \dots, K, \quad j = 1, \dots, J, \quad (12)$$

where θ_k denotes the average effect of the k -th treatment across all sub-populations and $\alpha_{k,j} = \theta_{k,j} - \theta_k$ is the treatment-by-stratum interaction term, representing the population-specific deviation of treatment k from the common effect θ_k (i.e., the departure from treatment effect homogeneity), where $\sum_{j=1}^J \alpha_{k,j} = 0$ for all $k = 1, \dots, K$.

To adopt a notation suitable for experiments evolving simultaneously under potentially dependent designs, we reintroduce the experimental indicator $\mathcal{S}$, by assuming that $\{\mathcal{S}_n\}_{n \geq 1}$ is an i.i.d. sequence ancillary for the likelihood, with $\Pr(\mathcal{S}_n = j) = \lambda_j > 0$ and $\sum_{j=1}^J \lambda_j = 1$, i.e., λ_j is the population proportion/prevalence of $\mathcal{P}_j$ over $\mathcal{P}$. Let $N_k(n_j) = \sum_{i=1}^{n_j} \delta_{ik} I_{\{\mathcal{S}_i=j\}}$ denote the number of patients assigned to treatment k in stratum j after n assignments and $Y_i = \sum_{j=1}^J \sum_{k=1}^K \delta_{ik} I_{\{\mathcal{S}_i=j\}} Y_i(k, j)$ be the observed outcome for patient i (where $I_{\{A\}}$ denotes the indicator of the event A).

Remark 2 Since $\theta_{k,j} = \mathbb{E}[Y_i | \delta_{ik} = 1, \mathcal{S}_i = j]$, ANOVA model (12) can be re-parameterized in terms of the marginal treatment effect $\tilde{\theta}_k$ in the overall population $\mathcal{P}$, namely $\tilde{\theta}_k = \mathbb{E}[Y_i | \delta_{ik} = 1] = \sum_{j=1}^J \lambda_j \theta_{k,j}$, leading to the constraint $\sum_{j=1}^J \lambda_j \alpha_{k,j} = 0$.

The inferential goal of $\tilde{E}$ consists in estimating/testing the collection of stratum-specific treatment effects $\{\theta_{k,j}\}_{\forall k,j}$, where the stratum specific MLE of $\theta_{k,j}$ is $\hat{\theta}_{k,n_j} = N_k(n_j)^{-1} \sum_{i=1}^{n_j} \delta_{ik} I_{\{\mathcal{S}_i=j\}} Y_i$. If the effect of the k -th treatment is homogeneous across

sub-populations, namely if $\alpha_{k,j} = 0$ for all $j = 1, \dots, J$, then $\theta_{k,j} = \theta_k$ and, analogously to (8), θ_k is estimated by a linear combination of the stratum-specific estimators with weights proportional to the corresponding sample sizes, namely by the pooled estimator $\hat{\theta}_n$ in (8).

This estimator combines the stratum-specific MLEs through a full pooling induced by homogeneity of the treatment effects. However, the degree of treatment heterogeneity across strata is typically unknown. Thus, we introduce a borrowing scheme that induces, for each treatment, a data-driven clustering of sub-experiments. Assuming that only experiments/strata exhibiting similar treatment effects should be aggregated, for each treatment k and stratum j , let

$$A_{k,j,n} = \left\{ r \neq j \in \{1, \dots, J\} : |\hat{\theta}_{k,n_r} - \hat{\theta}_{k,n_j}| \leq c_n \right\}$$

denote the set of sub-populations with similar estimated treatment effects at time n , where $\{c_n\}_{n \geq 1}$ is a positive decreasing sequence converging to zero as n increases. More generally, $|\hat{\theta}_{k,n_r} - \hat{\theta}_{k,n_j}|$ may be replaced by any scalar distance that consistently measures the discrepancies between the effects of each treatment across the sub-populations.

Within the overall experiment $\tilde{E}$, for each treatment k and stratum j information sharing across sub-populations is implemented through the borrowing estimator

$$\hat{\psi}_{k,j,n} = \frac{N_k(n_j) \hat{\theta}_{k,n_j} + \sum_{r \in A_{k,j,n}} N_k(n_r) \hat{\theta}_{k,n_r}}{N_k(n_j) + \sum_{r \in A_{k,j,n}} N_k(n_r)} = \frac{\sum_{r \in A_{k,j,n} \cup \{j\}} \sum_{i=1}^n \delta_{ik} I_{\{S_i=r\}} Y_i}{\sum_{r \in A_{k,j,n} \cup \{j\}} \sum_{i=1}^n \delta_{ik} I_{\{S_i=r\}}}, \quad (13)$$

that allows information sharing (in an adaptive manner) only across sub-experiments showing homogeneity. For each treatment k , $\hat{\psi}_{k,j,n}$ is a convex combination of $\hat{\theta}_{k,n_j}$ and the estimator obtained by pooling experiments in $A_{k,j,n}$ with weights given by the corresponding group size, i.e., $N_k(n_j) / \sum_{r \in A_{k,j,n} \cup \{j\}} N_k(n_r)$ and $\sum_{r \in A_{k,j,n}} N_k(n_r) / \sum_{r \in A_{k,j,n} \cup \{j\}} N_k(n_r)$, respectively.

For any experiment $j = 1, \dots, J$, define the vector of borrowing estimators across the K treatments as $\hat{\psi}_{j,n} = (\hat{\psi}_{1,j,n}, \dots, \hat{\psi}_{K,j,n})^t$. At the population level, the experiments to be aggregated to E^j are those identified by $A_{k,j}^* = \{r \neq j : \theta_{k,r} = \theta_{k,j}\}$, where for each of them $\psi_{k,j}^* = \theta_{k,j}$. Thus, the proposed borrowing mechanism is both treatment- and stratum-specific and accounts for the extreme situations of homogeneity and heterogeneity. When no similar strata are identified (i.e., in the case of heterogeneous treatment effects) $A_{k,j,n} = \emptyset$ and $\hat{\psi}_{k,j,n}$ coincides with the local estimator $\hat{\theta}_{k,n_j}$, while $\hat{\theta}_{k,n}$ is an overall estimator of the effect of treatment k obtained by averaging the stratum specific components $\hat{\theta}_{k,n} = \sum_{j=1}^J \omega(n_j) \hat{\psi}_{k,j,n} = \sum_{j=1}^J \omega(n_j) \hat{\theta}_{k,n_j}$ (see Remark 2). Whereas, in the case of complete homogeneity, the

borrowing mechanism automatically performs full pooling, recovering the most efficient estimator for the common treatment effect. When $A_{k,j,n} \neq \emptyset$, all the similar strata to j are aggregated and contribute to the estimator $\hat{\psi}_{k,j,n}$ and if there exist sub-populations with identical treatment effects, the sharing of information persists asymptotically within the corresponding cluster, leading to increased efficiency.

It is worth emphasizing that the approach presented here induces a sharp borrowing mechanism: strata are either fully aggregated when treatment effects are deemed homogeneous, or not aggregated at all. Such an approach may be too restrictive in some experimental settings, as it does not allow one to tune information sharing according to the degree of similarity between treatment effects. Extensions to borrowing mechanisms that weight information according to a measure of similarity between experiments represent a natural direction for future research.

5.1 Combining independent experiments in a heterogeneous population

If the experiments are independent (i.e., condition (7) holds), the results in Section 4.1 apply. The likelihood factorizes as before, stratum-specific estimators are independent, and both standard and borrowing estimators can be obtained by combining independent sources of information without additional inferential complications.

5.2 Combining dependent experiments in a heterogeneous population

When condition (7) does not hold, experiments are not independent and the allocation process in each sub-experiment depends on the information accrued in the others. In this case, the joint likelihood is as in Section 4.2. Depending on the nature of the information shared across experiments, two situations can be distinguished:

- if the allocation probabilities depend exclusively on the past treatment assignments (possibly collected across different experiments) the resulting design belongs to the class of covariate-adaptive designs; thus, the assignment process is ancillary to the likelihood and standard conditional inference given the realized allocation sequence remains valid;
- if the allocation probabilities depend on the responses observed in the other sub-experiments, the assignment process is no longer ancillary to the likelihood.

In practice, the interesting case for the present work is that of concurrent and interacting RA experiments evolving across strata. In the following, we establish consistency and asymptotic normality results for a class of CARA designs with outcome-dependent borrowing across concurrent experiments.

To incorporate information sharing at the design stage, the idea is to base the allocation probability of patient $(n + 1)$ belonging to stratum/experiment j to treatment

k on the borrowing estimator $\hat{\psi}_{k,j,n}$, namely assuming that

$$\Pr(\boldsymbol{\delta}_{n+1} = \mathbf{e}_k | \mathcal{F}_n, \mathcal{S}_{n+1} = j) = \frac{g(\hat{\psi}_{k,j,n})}{\sum_{\ell=1}^K g(\hat{\psi}_{\ell,j,n})}, \quad k = 1, \dots, K, \quad (14)$$

where $g(\cdot)$ is a continuous and monotonically increasing function (assuming a the larger the better scenario, accounting for ethical demand) with $0 < g(x) < \infty$ for every $x \in \mathbb{R}$. Note that this approach mirrors a type of interacting urns mechanism introduced in Aletti et al. (2024).

Since $\mathcal{S}_1, \dots, \mathcal{S}_n$ are iid, if we assume that each n_j diverges then by the law of large numbers the observed sampling proportions converge to the population prevalences, i.e., $\lim_{n \rightarrow \infty} \omega(n_j) = \omega_j^* = \lambda_j \in (0, 1)$ (otherwise, if n_j is bounded, then $\omega_j^* = 0$). This condition may fail in the presence of oversampling of specific subgroups, selective enrollment criteria, differential stopping across strata, or adaptive enrichment strategies. These scenarios fall outside the scope of the present work and will be the subject of future research. Let $\boldsymbol{\theta}_j = (\theta_{1,j}, \dots, \theta_{K,j})^t$ be the vector of treatment efficacy for stratum j and $\hat{\boldsymbol{\theta}}_{n_j}$ the corresponding estimator, the following theorem establishes strong consistency and asymptotic normality for both $\hat{\boldsymbol{\theta}}_{n_j}$ and $\hat{\psi}_{j,n}$ under (14).

Theorem 1 *Let $\{c_n\}_{n \geq 1}$ be a positive sequence of real numbers chosen such that $c_n \rightarrow 0$ and $\lim_{n \rightarrow \infty} c_n \sqrt{n / \ln(\ln n)} = \infty$. If $\lim_{n \rightarrow \infty} \omega(n_j) = \lambda_j \in (0, 1)$ for every sub-experiment $j = 1, \dots, J$, then*

$$\hat{\boldsymbol{\theta}}_{n_j} \rightarrow \boldsymbol{\theta}_j, \quad A_{k,j,n} \rightarrow A_{k,j}^*, \quad \hat{\psi}_{j,n} \rightarrow \theta_j \quad \text{and} \quad \boldsymbol{\pi}(n_j) \rightarrow \boldsymbol{\pi}_j^* = (\pi_{1,j}^*, \dots, \pi_{K,j}^*) \quad \text{a.s.}$$

where $\pi_{k,j}^* = \left[\sum_{\ell=1}^K g(\theta_{\ell,j}) \right]^{-1} g(\theta_{k,j})$ and therefore,

$$\boldsymbol{\pi}(n) \rightarrow \boldsymbol{\pi}^* = \sum_{j=1}^J \omega_j^* \boldsymbol{\pi}_j^*, \quad \text{and} \quad \boldsymbol{\rho}(n_j) \rightarrow \boldsymbol{\rho}_j^* = \text{diag}(\boldsymbol{\pi}^*)^{-1} \omega_j^* \boldsymbol{\pi}_j^* \quad \text{a.s.}$$

Moreover, by letting $\mathbf{V}_j = \text{diag}(\mathbb{V}(Y(k, j)))_{k=1, \dots, K}$, then

$$\text{diag}(\mathbf{N}(n_j))^{1/2} (\hat{\boldsymbol{\theta}}_{n_j} - \boldsymbol{\theta}_j) \xrightarrow{d} \mathcal{N}(\mathbf{0}_K, \mathbf{V}_j)$$

and the borrowing estimator is also asymptotically normal with effective sample sizes $\bar{N}_k(n_j) = \sum_{r \in A_{k,j}^* \cup \{j\}} N_k(n_r)$, namely

$$\text{diag}(\bar{\mathbf{N}}(n_j))^{1/2} (\hat{\psi}_{j,n} - \theta_j) \xrightarrow{d} \mathcal{N}(\mathbf{0}_K, \mathbf{V}_j),$$

where $\bar{\mathbf{N}}(n_j) = (\bar{N}_1(n_j), \dots, \bar{N}_K(n_j))^t$.

Proof The proof relies on adapting the arguments of Theorems 4.1 and 4.2 in Aletti et al. (2024) to the present interacting CARA framework. First, since $g(\cdot)$ is strictly positive and $\lambda_j > 0$, each treatment is assigned infinitely often within each stratum almost surely, so that $N_k(n_j) \rightarrow \infty$ a.s. This guarantees the strong consistency of the stratum-specific estimators $\hat{\boldsymbol{\theta}}_{n_j}$ (see Lemma A.3 in the Supplementary Material of Aletti et al. (2024)). The convergence

of the borrowing sets $A_{k,j,n}$ follows from the condition $c_n \sqrt{n/\ln \ln n} \rightarrow \infty$, which is motivated in the proof of Theorem 4.2. Then almost sure convergence of $\hat{\psi}_{j,n}$ follows from the consistency of the stratum-specific estimators. Finally, asymptotic normality is obtained by applying the martingale central limit theorem, as in the proof of Theorem 4.2 of [Aletti et al. \(2024\)](#). $\square$

When no other stratum shares the same treatment effect, i.e. $A_{k,j}^* = \emptyset$, then $\hat{\psi}_{k,j,n}$ coincides with $\hat{\theta}_{k,n_j}$ and the effective sample size is $N_k(n_j)$. Conversely, when $A_{k,j}^* \neq \emptyset$, borrowing persists asymptotically within the corresponding cluster of strata/experiments. In this case, the estimator $\hat{\psi}_{k,j,n}$ effectively aggregates information from all strata in $A_{k,j}^* \cup \{j\}$ and the effective sample size is then $\bar{N}_k(n_j) = N_k(n_j) + \sum_{r \in A_{k,j}^*} N_k(n_r)$, yielding a faster rate of convergence.

Example 1 (Illustration of Theorem 1) Consider a two-arm trial ($K = 2$) conducted in $J = 3$ sub-populations. Assume binary responses, so that $Y(k, j) \sim \text{Bernoulli}(\theta_{k,j})$, for $k = 1, 2$ and $j = 1, 2, 3$. Suppose that $\theta_1 = \theta_2 = (0.65, 0.4)^t$ and $\theta_3 = (0.5, 0.4)^t$, then treatment 1 has the same effect in strata 1 and 2 but a different effect in stratum 3, whereas treatment 2 is homogeneous across all the strata. Therefore,

$$A_{1,1}^* = \{2\}, \quad A_{1,2}^* = \{1\}, \quad A_{1,3}^* = \emptyset,$$

while

$$A_{2,1}^* = \{2, 3\}, \quad A_{2,2}^* = \{1, 3\}, \quad A_{2,3}^* = \{1, 2\}.$$

By setting $g(x) = (1 - x)^{-1}$ in (14), the stratum-specific allocation proportions converge to

$$\pi_1^* = \pi_2^* = (0.632, 0.368)^t \quad \text{and} \quad \pi_3^* = (0.545, 0.455)^t.$$

Although treatment 2 has the same efficacy in all the strata, the asymptotic allocation proportions to the two treatments are different from strata 1-2 and 3 (since the relative performance of treatment 2 with respect to treatment 1 changes). Hence, assuming 100 patients are observed in each stratum, under the limiting allocation proportions the number of subjects assigned to each treatment is reported in Table 1.

Table 1 Number of assignments for each stratum and treatment

	Treatment 1	Treatment 2	Tot.
Stratum 1	$N_1(n_1) = 63$	$N_2(n_1) = 37$	$N(n_1) = 100$
Stratum 2	$N_1(n_2) = 63$	$N_2(n_2) = 37$	$N(n_2) = 100$
Stratum 3	$N_1(n_3) = 55$	$N_2(n_3) = 45$	$N(n_3) = 100$

With regard to treatment 1, the stratum-specific estimators are based on approximately 63, 63, and 55 observations, respectively. However, since strata 1 and 2 are homogeneous, the corresponding borrowing estimators $\hat{\psi}_{1,1,n}$ and $\hat{\psi}_{1,2,n}$ asymptotically combine the information from both strata, so that their effective sample sizes become $\bar{N}_1(n_1) = \bar{N}_1(n_2) = 126$. With regard to stratum 3, no asymptotic borrowing occurs for treatment 1 and therefore $\hat{\psi}_{1,3,n}$ coincides with the corresponding stratum specific estimator with effective sample size

$N_1(n_3) = 55$. Therefore, for treatment 1, $\hat{\psi}_{1,1,n}$ and $\hat{\psi}_{1,2,n}$ are based on roughly twice as many observations as the corresponding stratum-specific estimators.

For treatment 2, all strata are asymptotically identified as homogeneous, so that the borrowing estimators $\hat{\psi}_{2,1,n}$, $\hat{\psi}_{2,2,n}$ and $\hat{\psi}_{2,3,n}$ asymptotically pool information across the three strata and are based on the combined sample size $\bar{N}_2(n_1) = \bar{N}_2(n_2) = \bar{N}_2(n_3) = 119$. This corresponds to the case of maximal asymptotic benefit from borrowing.

In practice, during the sequential procedure, at each stage n the sets $A_{k,j,n}$ are obtained by comparing the current stratum-specific estimators through the threshold c_n . Thus, borrowing is entirely data-driven not only at the analysis stage but also in the design phase, where subjects are allocated to the treatments on the basis of the borrowing estimators. With Theorem 1 we show that the sets $A_{k,j,n}$ eventually coincide almost surely with the population sets $A_{k,j}^*$, so that asymptotically the procedure identifies the correct homogeneous clusters and borrows information only within them.

When the discrepancy between the stratum-specific effects of a given treatment is fixed, albeit modest, sharing information across strata may still be meaningful, since the stratum effect is limited. The degree of information sharing changes according to the choice of c_n (i.e., on its rate of convergence to zero); hence, even for large sample sizes, the procedure may still perform borrowing across strata. However, Theorem 1 guarantees that borrowing eventually ceases and both the stratum specific and the borrowing estimators remain asymptotically normal. These results do not hold when the discrepancy between the stratum specific treatment effect is not assumed to be fixed but depends on n (as in Pitman's framework of hypothesis testing with local alternatives); see Section 6 for an in depth discussion on this point.

The theoretical results developed in this Section rely on the assumption that each sub-experiment is sampled infinitely often. If some n_j s are bounded or a stopping rule is introduced that leads to the permanent cessation of recruitment in a given stratum, the corresponding treatment effects can no longer be consistently estimated, as the amount of information available for that stratum remains finite. In this case, asymptotic results for the stopped stratum, such as consistency and asymptotic normality, do not apply and also $A_{k,j,n} \not\rightarrow A_{k,j}^*$ almost surely. From the perspective of the borrowing mechanism, the estimates associated with a bounded sub-experiment become frozen after its maximum sample size is reached. Consequently, allowing such a stratum to continue contributing to information borrowing would introduce bias or instability in the estimation of treatment effects for the remaining strata. For this reason, these sub-experiments should be excluded from the borrowing sets once recruitment stops.

6 Discussion

This paper investigates the asymptotic properties of estimators obtained from combining multiple adaptive experiments intended to compare two or more treatments, opening the way for several potential follow-ups. We focus on settings in which the adaptation process affects the probabilities of treatment assignments, without considering stopping rules, which are nonetheless an integral part of many designs. While the inclusion of interim decisions for early stopping is relatively straightforward in

the case of independent experiments, the analysis becomes more challenging when experiments depend on one another. This occurs particularly for informative (i.e., Response-Adaptive) stopping mechanisms, for instance in multistage experiments with interim analyses, where sample size recalculation and inference at a given stage depend on previously observed outcomes; in such settings, stable convergence tools can be applied.

More generally, in the presence of complex non-ancillary adaptive mechanisms (like, e.g., group-sequential decisions, sample size re-estimation, enrichment designs), the effect of adaptation may persist asymptotically and likelihood-based inference comes with information loss; the arguments developed in this paper are no longer sufficient, since equation (2) does not correspond to the conditional likelihood and condition (5) may fail to hold, either because the limiting allocation proportion is random, or because one or more of its components converge to 0 or 1. As a consequence, non-standard large sample inference (e.g., information loss, bias, asymptotic non-normality of the MLEs) may occur (see [Tarima and Flournoy \(2024, 2019\)](#); [Flournoy and Tarima \(2023\)](#); [Lane et al. \(2014\)](#)). Such broader classes of adaptive designs are highly relevant in practice and require different theoretical tools, especially in CARA settings, where stratum-specific stopping rules or subgroup-specific adaptations may fundamentally alter the joint asymptotic behavior of the estimators. In such a case, the functional Central Limit Theorem for sequential monitoring, combined with borrowing schemes (see [Aletti et al. \(2024\)](#)), may help achieve high precision even with moderate sample sizes, thereby reducing the overall size and duration of the experiment.

Another important issue concerns the amount of borrowing that should be used in the different scenarios. For instance, in the approach discussed in Section 5, information is either borrowed or not borrowed on the basis of a measure of homogeneity among sub-experiments. Although this strategy provides conceptual clarity and strong theoretical guarantees, it may be overly restrictive in some applications:

- (i) the concept of homogeneity can be formalized/measured in many different ways. For example, it may be based on the treatment-by-stratum interaction term, which represents the population-specific deviation of a given treatment from the common effect. This alternative definition may allow for less rigid aggregation criteria. More flexible borrowing schemes could also be considered: for instance, when historical data are available, previous observations could be aggregated in the current study with weights decreasing as the temporal distance between experiments increases. Such approaches would allow partial borrowing even in the presence of moderate discrepancies across studies, potentially improving efficiency while controlling bias;
- (ii) while Theorem 1 provides a solid asymptotic framework for information borrowing, when small differences between the stratum-specific effects are present, both the sample size and the choice of c_n play a crucial role. Although for fixed discrepancies the proposed procedure borrows information correctly in the asymptotic regime and the estimators are asymptotically normal, if the actual discrepancy is of order $o(1/\sqrt{n})$ - as in Pitman's framework of hypothesis testing with local

alternatives - the effect of the tuning sequence c_n may be neutralized; therefore, asymptotic normality is no longer guaranteed and the proposed procedure could induce a considerable increase in MSE compared with conventional procedures that do not aggregate data. Therefore, while asymptotic guarantees are a necessary requirement, a complete assessment of the practical use of borrowing should also examine local discrepancies of infinitesimal order, where information sharing may be particularly uncertain and, in the limit, potentially harmful.

Exploring these extensions represents a promising direction for future research.

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